

ENGINEERING PORTFOLIO · 2026

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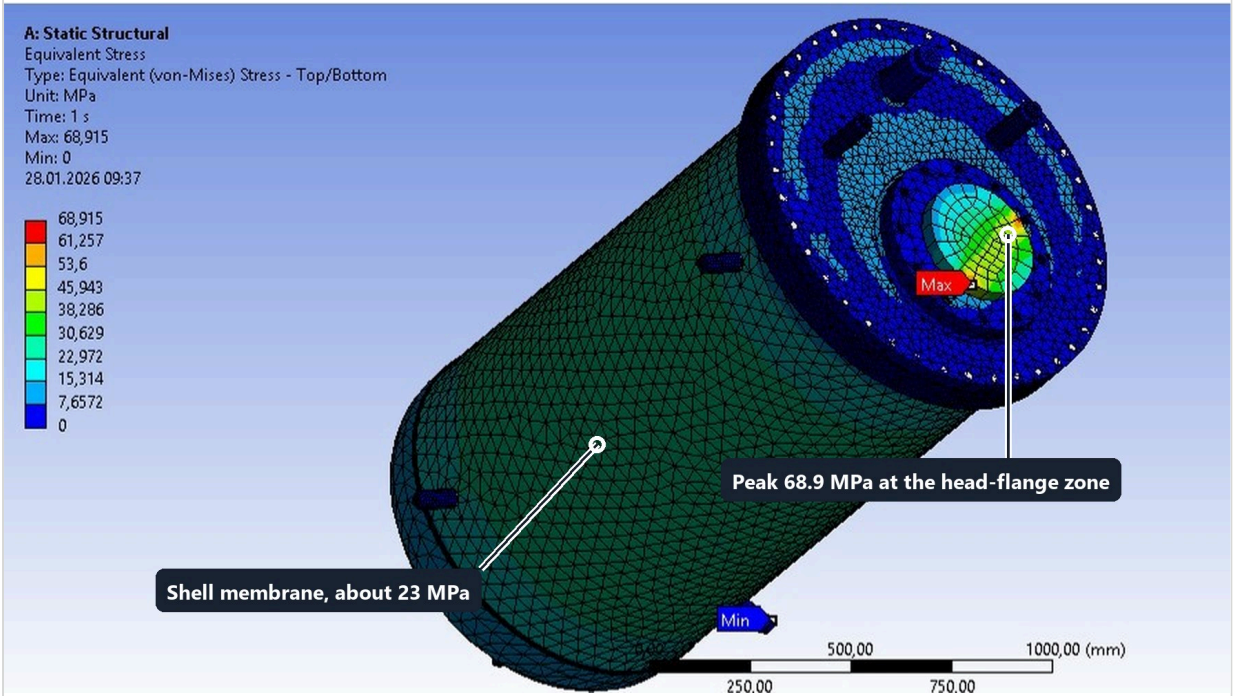
01 · PRESSURE VESSEL · STRUCTURAL ANALYSIS

# Shell theory said 11.2. The real head said 3.05.

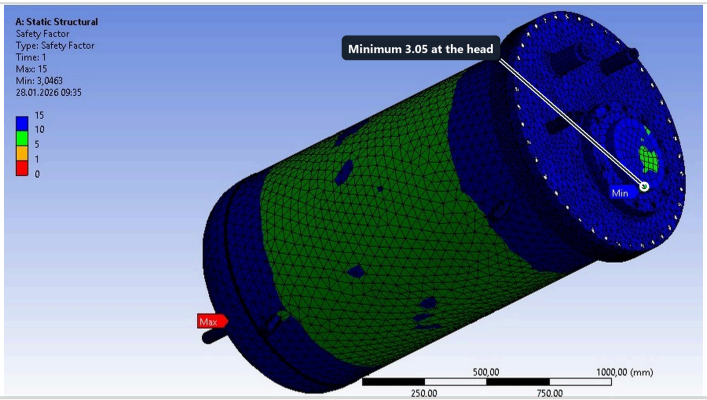
Mechanical System Design project, İÜC · Oct 2025 - Jan 2026
Advisor: Dr. Murat Bozkurt · individual work

I calculated the stresses of a pressure vessel by hand, built an ANSYS model of the vessel and its bolted head and showed that the head-flange region, not the shell, sets the safety margin.

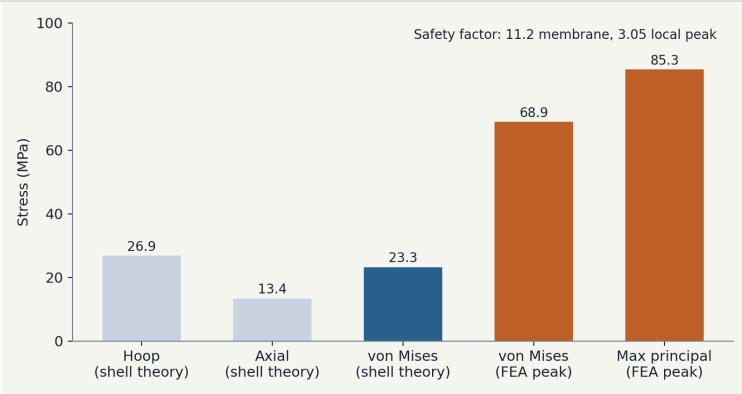
| 3.05 | 11.2 | 68.9 MPa | 45.8 m |
|---|---|---|---|
| minimum safety factor (FEA, at the head) | safety factor of the shell membrane (theory) | peak von Mises stress | worst-case blast safety distance |




Equivalent stress, ANSYS Mechanical 2022 R1




Safety factor map




Shell theory against FEA peak values

01 · PRESSURE VESSEL · METHOD

# Two routes, compared at the right level

1 · HAND CALCULATION
Hoop, axial and von Mises stress, radial growth

→

2 · FEA MODEL
Vessel and bolted head, linear static, refined mesh

→

3 · TWO-LEVEL CHECK
Membrane vs theory, peaks at discontinuities

→

4 · CONSEQUENCE
Stored energy, TNT equivalent, safety distance

## PROBLEM

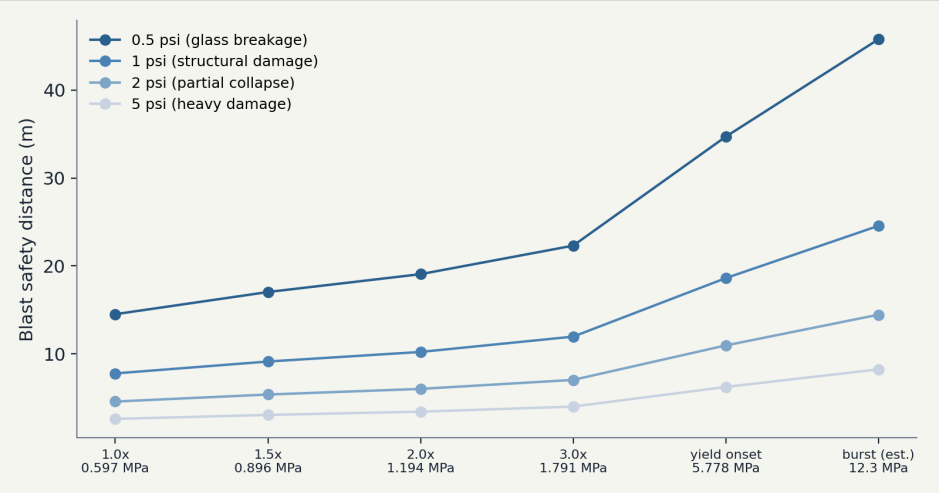
Shell r = 450 mm, t = 10 mm, head 12 mm, L = 1600 mm, SA-516 Gr.70 (yield 260 MPa), internal pressure 0.597 MPa.

## WHAT I DID

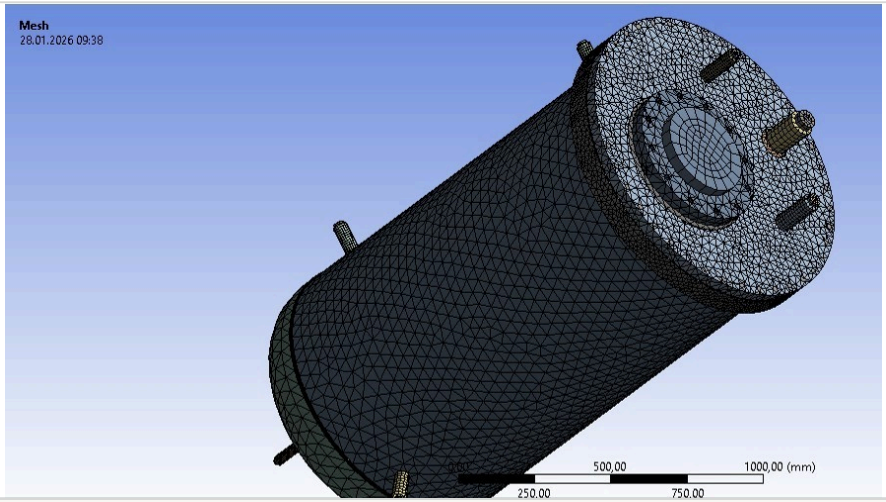
- **Calculated** the shell stresses with thin-wall theory: von Mises 23.3 MPa, radial growth 0.051 mm.
- **Modelled** the vessel and bolted head in ANSYS Mechanical with mesh refinement at the flange and head.
- **Compared** theory with the shell membrane and read the FEA peaks as bending at discontinuities, instead of forcing one number onto both.
- **Extended** the study with stored energy (isentropic gas), TNT equivalent and Hopkinson-Cranz blast distances. Article in preparation.

## WHAT I WOULD DO NEXT

A mesh independence study at the peak and a minimum-constraint support instead of the fixed base.




Blast safety distance by pressure scenario (V = 1.1 m³)




Mesh, refined at the discontinuities

02 · TEKNOFEST UAV · DESIGN AND MANUFACTURING

# A landing gear 50% under its weight target

TEKNOFEST International UAV Competition · Aug 2024 - Aug 2025
Team project · I designed the landing gear

For a 7 kg quadcopter I designed a skid landing gear from carbon-fibre tubes and 3D-printed joints, checked it with static FEA and brought it in 50% below the specification weight target.

**50%**
below the specification weight target

**7.05 kg**
take-off mass of the UAV

**292 mm**
landing gear height

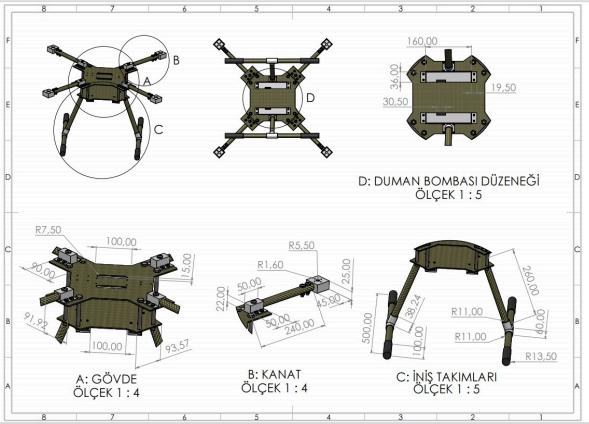
**415 mm**
landing gear width



The whole system: UAV assembly in SolidWorks



My subsystem: skid legs, carbon tubes, printed T-joints




Drawing sheet with gear dimensions

# Light, stiff and easy to fit

**1 · REQUIREMENTS**
7 kg take-off mass, camera clearance, hard landings

→

**2 · PARAMETRIC CAD**
Skid legs, T-joints and flanges in SolidWorks

→

**3 · STATIC FEA**
Landing loads on the gear in ANSYS

→

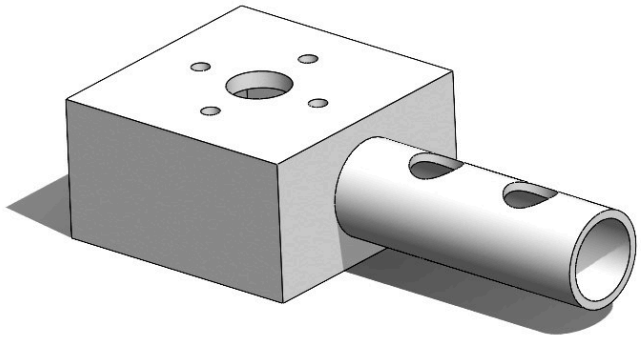
**4 · BUILD**
Carbon tubes, FDM joints in PLA, ABS and ASA

## DESIGN DECISIONS

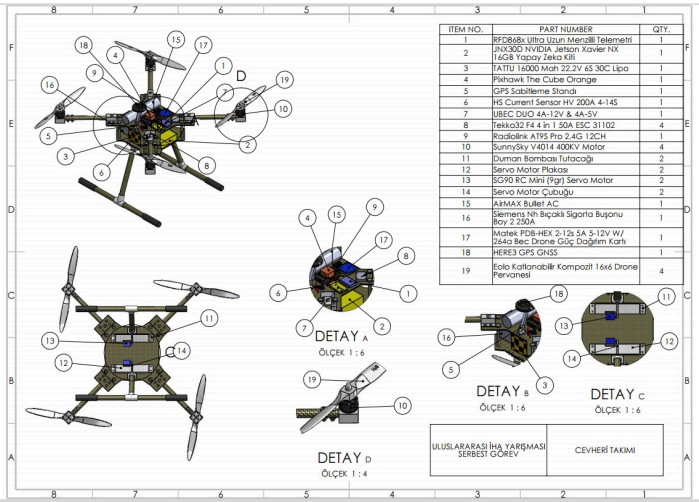
- **Skid gear** instead of arm-mounted legs: out of the camera's view and spreads the landing load over a larger area.
- **Carbon-fibre tubes** with a 5 mm wall on the legs, thicker than the arms, from a local manufacturer.
- **3D-printed T-joints and angled flanges** in ABS for strength-to-weight and easy printing.
- **Rubber pads** on the skids to soften the landing shock.
- **Bolted to the frame** like the other parts, for easy assembly.

## SKILLS

Parametric CAD · Static FEA · FDM printing · Material selection · Drawings



3D-printed joint detail




Assembly drawing with bill of materials

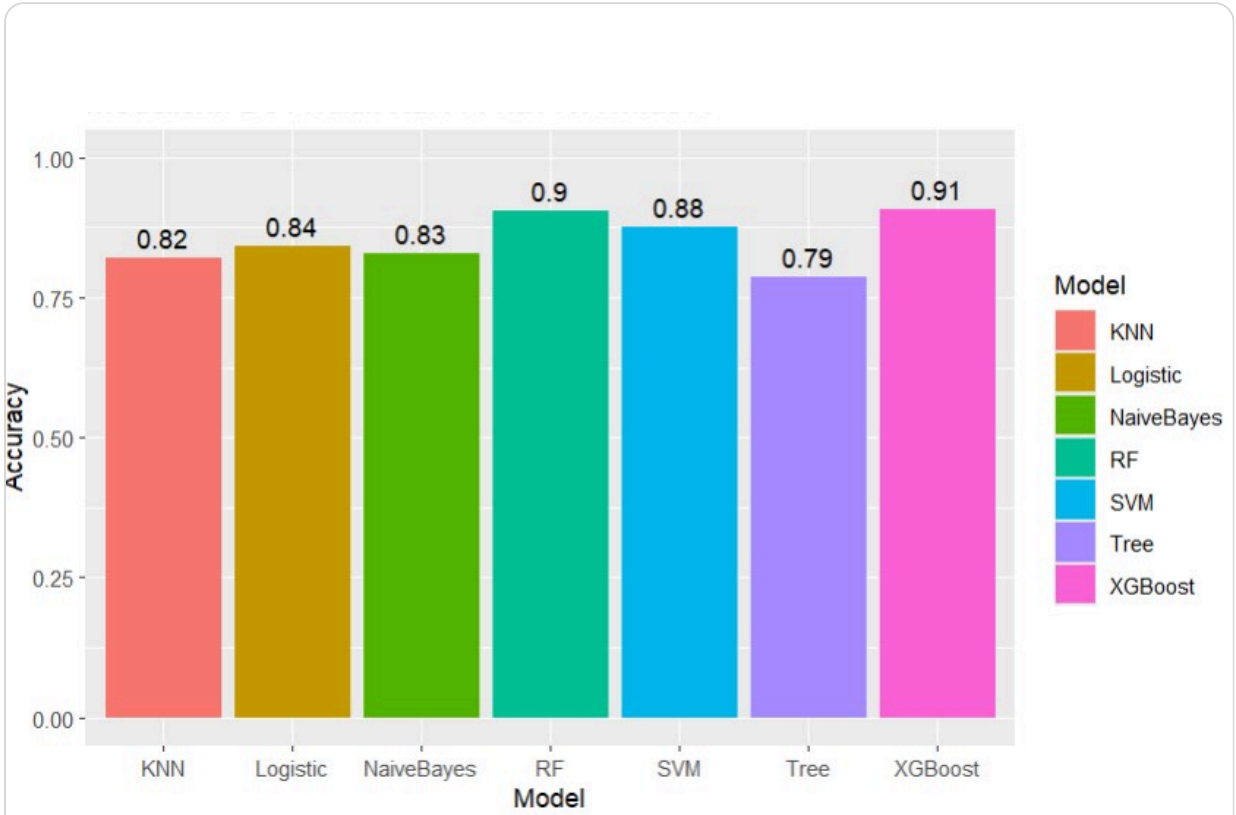
03 · MACHINE LEARNING · QUALITY

# Telling steel plate defects apart at 91.3%

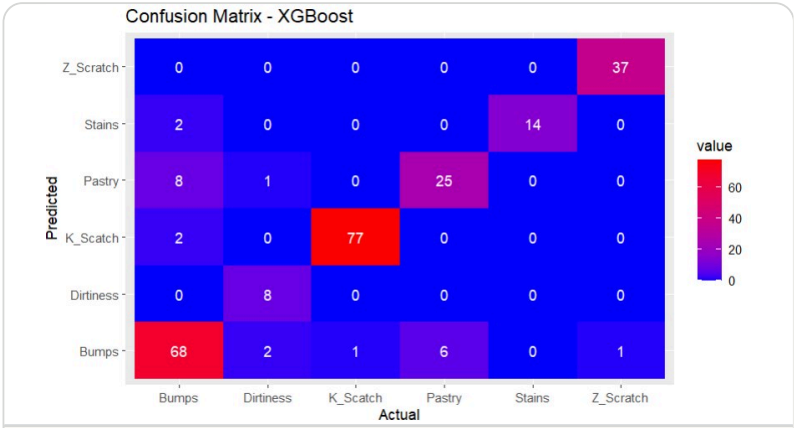
Machine learning course project · Feb 2025 - Jun 2025
Individual work · R

I compared seven classifiers on steel plate inspection data and combined the best three into a voting ensemble that identifies the fault type with 91.3% accuracy.

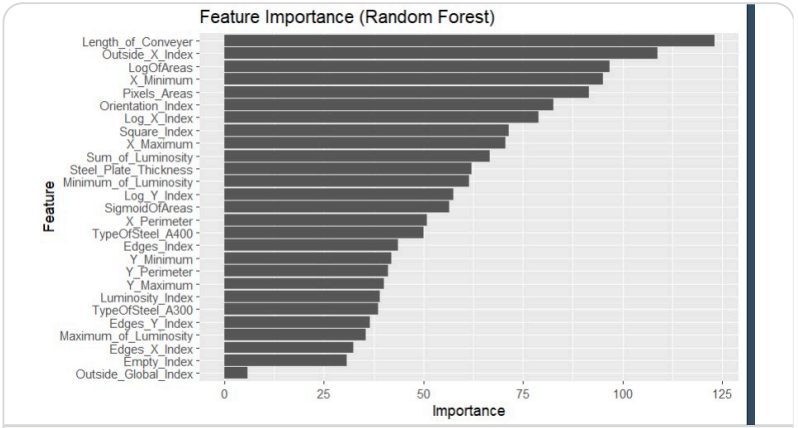
| 91.3% | 0.895 | 7 | 27 |
|---|---|---|---|
| voting ensemble accuracy | macro F1-score | models compared | input features |




Test accuracy of the seven models




Confusion matrix (XGBoost)




Random forest feature importance

03 · MACHINE LEARNING · METHOD

# From raw measurements to a voting ensemble

**1 · DATA**
UCI Steel Plates Faults, 27 features per plate

→

**2 · PREPARATION**
80/20 split, scaling, oversampling, Boruta selection

→

**3 · SEVEN MODELS**
From logistic regression to XGBoost

→

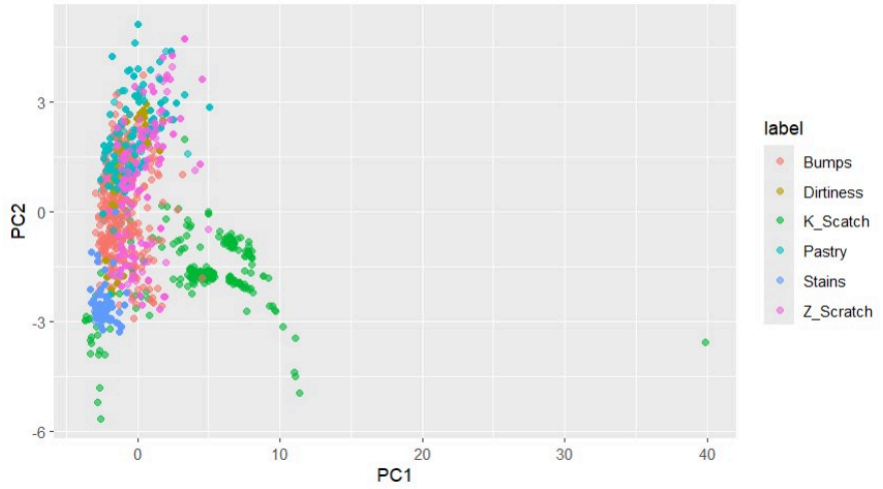
**4 · ENSEMBLE**
Majority vote of XGBoost, random forest and SVM

## WHAT I DID

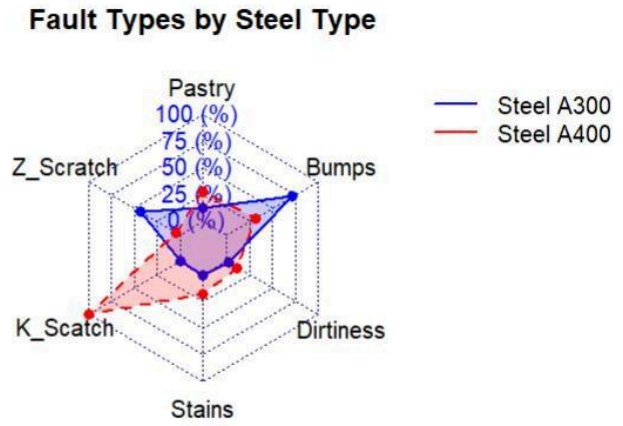
- **Cleaned** the data, left out the ambiguous "other faults" class and balanced the minority classes by oversampling.
- **Selected** the informative features with Boruta.
- **Compared** logistic regression, decision tree, random forest, k-NN, naive Bayes, SVM (5-fold CV for C) and XGBoost.
- **Combined** XGBoost (91%), random forest (90%) and SVM (88%) by majority vote: 91.3%, precision 0.909, recall 0.891.

## WHAT IT SAYS ABOUT THE PROCESS

K-scratch and bumps are recognised well, Z-scratch is often confused. Conveyor length ranks highest, so the model leans on line and geometry measurements.




PCA view: K-scratch separates, other faults overlap




Fault profile by steel type (A300 vs A400)

04 · MORE WORK

# Formula Student

İÜC RACING FORMULA STUDENT · POWERTRAIN TEAM · 2023 - 2026

**Designed** the chain guard in SolidWorks and verified its strength with static analysis in ANSYS. Revised the bracket design for a **15% weight reduction**.

SolidWorks | Static FEA | Lightweighting

## INTERNSHIPS

- **FESTO**, 2025: traced mechanical, pneumatic, electrical and software faults in a robotic system and recommissioned it.
- **BEKO**, manufacturing, 2025: found loss sources at 5 critical assembly stations and proposed 2 improvements.
- **Midas Makina**, design, 2024: revised SolidWorks models and prepared manufacturing drawings.
- **Yılmaz Redüktör**, workshop, 2024: machining, heat treatment and gear quality control on site.

## SKILLS

SolidWorks (CSWP), ANSYS Mechanical, AutoCAD, Siemens NX, CATIA, Fusion 360, MATLAB/Simulink, R, MS Excel. Static and modal FEA, topology optimisation, GD&T, design for manufacturability, injection moulding and sheet metal design, root cause analysis, lean manufacturing.

## CONTACT

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